Symmetric Fixed Points of the Ricci Flow

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Midterm Presentation
October 15, 2026

Introduction

- Ricci flow is a process that smooths out curved shapes over time, but it can sometimes cause them to collapse into "singularities".
- To fix these singularities, we use special shapes called Expanding Ricci Solitons that grow in a perfectly uniform, self-similar way.
- They are defined by a metric g and a potential function f satisfying: $\mathrm{Ric} + \nabla^2 \mathrm{f} + \mathrm{g} = 0$.
- We are studying three specific families of these fixed shapes: $\mathrm{R}^3 \times \mathrm{S}^1$, $\mathrm{R}^2 \times \mathrm{S}^2$, and R^n (Bryant solitons).

Project Goals

- Goal: Explore and prove how these fixed shapes grow and behave and study the properties as we go to spatial infinity (i.e. what the shapes/spaces look like as our distance from a fixed reference points becomes very large).
- Question 1: What shape do these objects approach as they grow infinitely large? (Their "asymptotic cone")
- Question 2: How do the starting conditions dictate the final, infinite-size shape?
- Question 3: For Bryant solitons, how does the starting state exactly determine the final shape?

Governing ODEs

- By assuming the shapes have high symmetry, we can reduce the complex math into a system of Ordinary Differential Equations (ODEs):

$$a'' = -2\frac{a'b'}{b} + a'f' + a$$

$$b'' = \frac{1-(b')^2}{b} - \frac{a'b'}{a} + b'f' + b$$

$$f'' = \frac{a''}{a} + 2\frac{b''}{b} - 1$$

- Challenge: These equations require dividing by variables that start at zero ($b = 0$). This creates a mathematical "division-by-zero" error right at the starting line ($r = 0$).

Computer Assisted Proof

- Goal: We want to deduce from our numerics the genuine existence of expanders asymptotic to cones with $R > 0$.
- $r = \epsilon \to r = 0.01$: Found max error of Taylor Series expansion from 11th term and beyond.
- $r = 0.01 \to r = 20$: Used a Julia script to bound errors for a, a', b, b', f, f' from $r = 0.01$ to $r = 20$.
- $r = 20 \to r = \infty$: Define Lyapunov Ratios to verify that the space expands smoothly without singularity for all $r \in [0, \infty)$.

Results: Mapping the Asymptotic Cones (3D View)

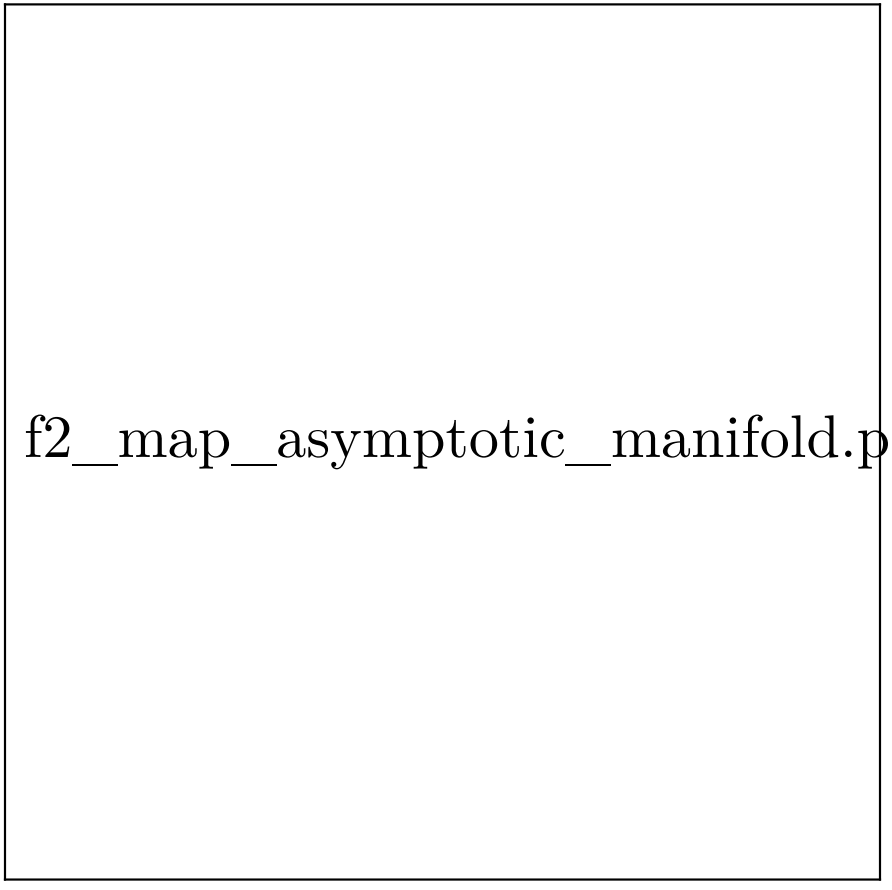


Figure: Graphed b_0 on the z-axis with a'_∞ and b'_∞ on the x and y axes and $f''(0)$ color-scaled.

Results: Achievable Cone Limits (2D Projection)

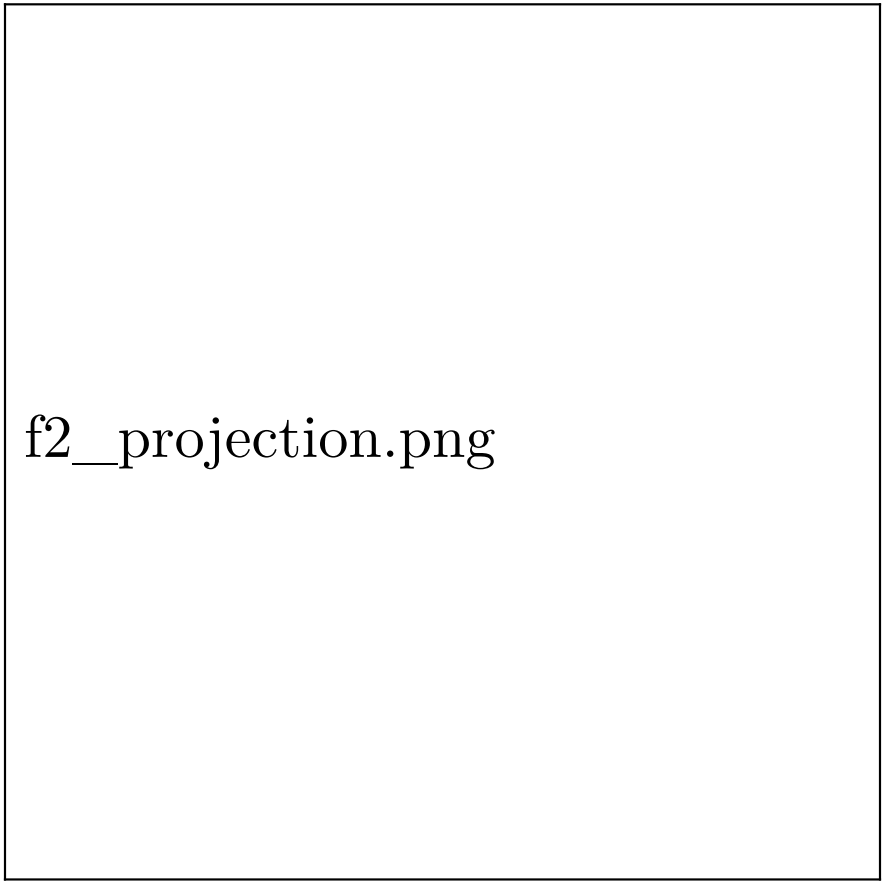


Figure: Projected previous image onto a'_∞ and b'_∞ as x and y axes.

Results: Bryant Soliton Thresholds

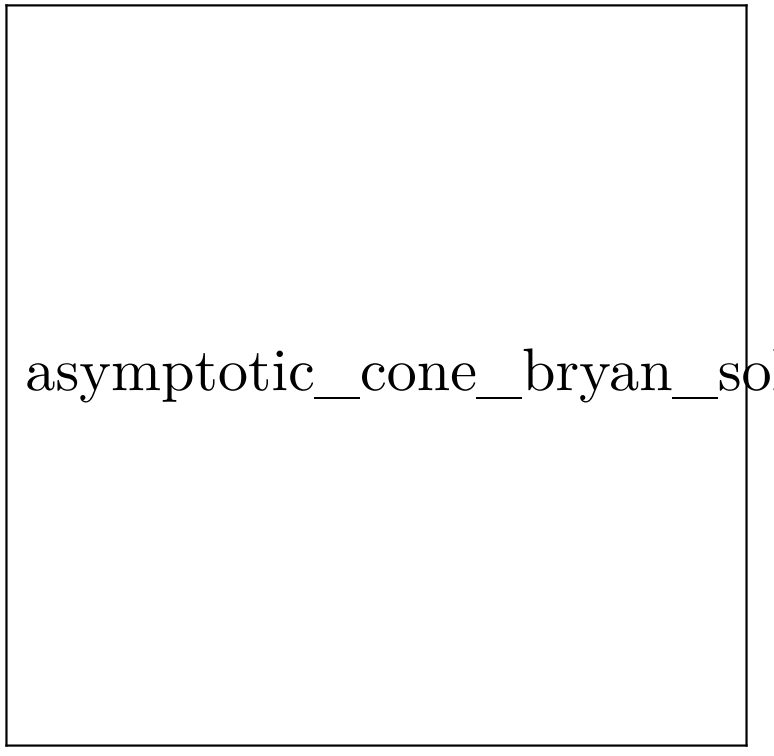


Figure: For Case 2.4, we plotted $\lim_{r\to\infty} \sqrt{h(r)}$ on the y axis and $\mu_1 = h'(0)$ on the x axis.

Next Steps

- Estimate the relative entropies of expanders, defined by Deruelle-Schulze, asymptotic to the same cone", where relative entropy has been defined.